

Akashi Kaikyo Bridge

An extraordinary feat of Japanese engineering, the world's longest suspension bridge



Opened on April 5, 1998, the Akashi Kaikyo Bridge—also called the Akashi Strait Bridge—carries a six-lane highway from Kobe city on Honshu Island to Iwaya on Awaji Island. The bridge is part of a larger system—the Honshu–Shikoku bridge project—which links the islands of Honshu and Shikoku across Japan's Inland Sea. The bridge is not only the world's longest suspension bridge, extending a total of 12,831 ft (3,911 m), but also one of the tallest, its two towers reaching 975 ft (297 m) into the sky.

Overcoming adversity

Spanning the Akashi Strait—a busy shipping route—was a considerable engineering challenge, not least because the area regularly experiences some of the world's worst storms and is seismically unstable. Indeed, while the bridge was under construction, an earthquake forced two of its towers 3 ft (1 m) farther apart. The designers used an innovative system of steel-truss girders and devices called tuned mass dampers (TMDs) to counteract the effects of wind and earthquakes. The triangular steel girders make the bridge rigid but also allow wind to pass through, while the TMDs swing in the opposite direction to the wind, balancing the bridge and cancelling sway. The bridge is capable of expanding and contracting by several feet. It can also withstand 180-mph (290-kph) winds and earthquakes of magnitude 8.5.

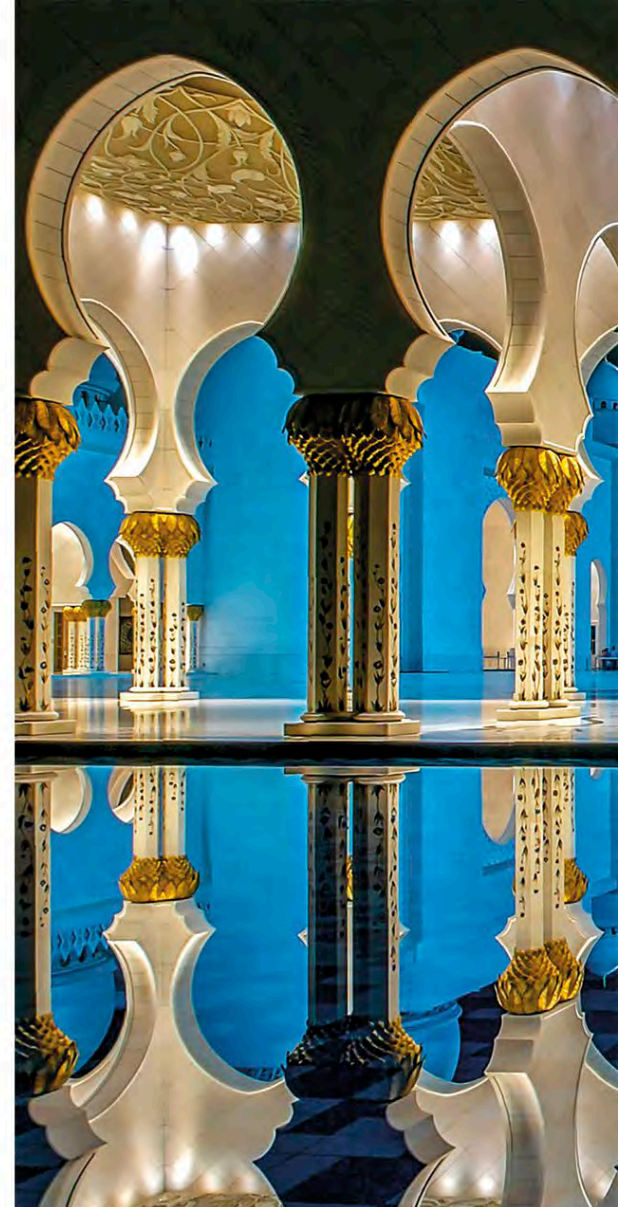


△ VIEW FROM BELOW

A steel-and-glass tunnel under a section of the bridge provides visitors with an unusual view of the scale of the construction.

▽ LONGEST SUSPENSION BRIDGE

Triangular steel girders provide the Akashi Kaikyo Bridge with the stability needed to span such an astonishing distance across the treacherous strait between the Honshu and Awaji islands.



△ REFLECTED BEAUTY

Pools of water surround the mosque's courtyard, reflecting the gilded and painted columns and elegant ceilings of its arcades. The mosque is illuminated at night in accordance with the phases of the Moon.

HYPOSTYLE MOSQUE

The Sheikh Zayed Mosque is a model hypostyle mosque (in which the prayer hall is made up of rows of columns). It has a main hall with 96 pillars, two open prayer halls, and a colonnaded courtyard. Four minarets, each 348 ft (106 m) high, sit at the corners of the courtyard.

